XIAN JIE LOH

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Available for a 24-week credit-bearing internship (NUS ATAP)

Internship Period: 4 Jan 2027 – 18 Jun 2027

xjie97.github.io/#work

EDUCATION

National University of Singapore Aug 2023 - Aug 2027 (expected)

Bachelor of Computing in Computer Science

Relevant coursework: Software Engineering; Data Structures and Algorithms; Computer Organisation; Computer Networks Practice; Computer Security.

Nanyang Polytechnic Apr 2015 - May 2018

Digital and Precision Engineering | Injection Moulding specialisation

TECHNICAL SKILLS

Languages: Python, Java 17, C, JavaScript; TypeScript for web prototyping

Computer vision: MediaPipe, OpenCV, NumPy; pose estimation, video data preparation

Applications: React Native, Expo, React, Next.js, JavaFX, Firebase, AsyncStorage

AI-assisted development: Prompt engineering, rapid prototyping, debugging, unit testing

Software engineering: System design, automation, regression testing, system validation

Tools and delivery: Git, GitHub, Gradle, JUnit 5, CI, PlantUML, Figma, Vercel

Collaboration: Cross-functional teamwork, stakeholder communication, process improvement

PROJECT EXPERIENCE

SquatVision | Personal Computer Vision Project (WIP) Oct 2026 - Present

Python, MediaPipe, OpenCV, NumPy, unittest, Git | GitHub

- Built a Python pipeline for recorded squat videos: pose estimation, hip/knee angles, repetition counting, movement-phase timestamps, bottom-pause estimation and video/JSON/CSV exports.
- Investigated segmentation, image boundaries, optical flow and temporal checks for anatomical depth references; reject unreliable measurements as UNAVAILABLE. Automatic IPF depth judging remains WIP.
- Used AI-assisted development for prototyping, debugging and regression-test development; added manual annotation and diagnostic review tools to support validation against source footage.
- Applied powerlifting coaching knowledge to computer vision and product scoping; prioritised squat-first refinement before planned bench/deadlift analysis and Android/iOS video import.

EduConnect | Software Engineering Team Project Oct - Nov 2025

Java 17, JavaFX, JUnit 5, Gradle, Git, PlantUML | GitHub

- Implemented 2 lookup commands to retrieve contact details and weekday tuition sessions; added chronological sorting with student-name tie-breaks for predictable schedule review.
- Added 6 test classes for lookup commands, parsers and formatting to check valid and invalid inputs; documented the features with UML diagrams and user guides for maintenance.

Tkit | Individual Software Engineering Project Aug - Oct 2025

Java 17, JavaFX, JUnit, Gradle, Checkstyle | GitHub

- Built a task manager supporting 10 commands and 3 task types; reused parsing, domain and storage modules across 2 interfaces, CLI and JavaFX, to avoid duplicating core logic.
- Protected saved tasks with atomic file replacement and malformed-record handling; added 11 test classes covering commands, date parsing, task models and storage.

Progress Pal | NUS Orbital Project May - Jul 2024

JavaScript, React Native, Expo, Firebase, AsyncStorage

- Developed a strength tracking application for 2 mobile platforms, Android and iOS, using one React Native codebase for workout planning, exercise logging and training history.
- Integrated Firebase authentication and persistent local storage to retain records between sessions; delivered the project at NUS Orbital's Apollo 11 (Advanced) level.

Exchango | Interaction Design Course Project 2026

Figma, React, Next.js, TypeScript, Vercel | Portfolio | Case study | GitHub

- Designed 2 core user journeys for exchange students, hosting trips and finding travel partners; resolved a gap identified in design feedback by adding a trip-creation entry point.
- Published a 10-section responsive web case study covering research, design iterations and an embedded prototype, allowing reviewers to inspect the proposed experience and rationale.

WORK EXPERIENCE

ETLA Limited | Assistant NPI Engineer Aug 2018 - Feb 2023

Applied skills: SAP, process planning, engineering change coordination, quality documentation

- Qualified manufacturing processes for 10+ medical and semiconductor components per project, translating drawing tolerances into production and inspection requirements to support part acceptance.
- Managed SAP-based coordination across engineering, production and suppliers to track schedules and resolve dependencies; aligned engineering changes with production and delivery requirements.
- Translated customer blueprints into process plans across 4 stages: machining, welding, finishing and quality control; incorporated welding qualification requirements under ISO 15614-11.

LEADERSHIP AND ACTIVITIES

Helix Gym | Founder Sep 2024 - May 2026

- Founded a strength training community of 50+ residents; established weekly sessions and beginner onboarding, using Microsoft Forms and Telegram to organise participation and communication.

Helix House Orientation | Corporate Relations Executive Feb - Aug 2026

- Negotiated vendor sponsorship support for orientation activities serving a residence of 500+ students; coordinated sponsored-item logistics against event deadlines and venue requirements.

NUS Powerlifting | Student Coach Oct 2023 - Aug 2024

- Reviewed athletes' exercise technique and performance data to identify coaching priorities; adapted training guidance and programming to individual development needs.

ADDITIONAL INFO

Interests: Strength training, cooking, and narrative and strategy games.